Introduction

Benoît Legat

Understand simple things deeply

☐ Full Width Mode ☐ Present Mode

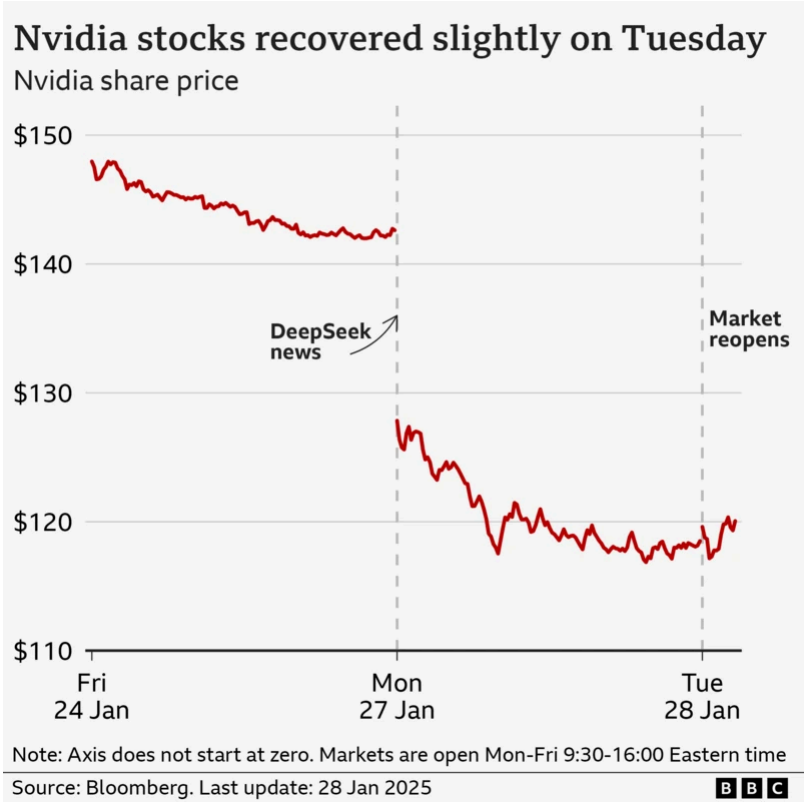
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Distillation

2.3.1. *Cold Start*

Unlike DeepSeek-R1-Zero, to prevent the early unstable cold start phase of RL training from the base model, for DeepSeek-R1 we construct and collect a small amount of long CoT data to fine-tune the model as the initial RL actor. To collect such data, we have explored several approaches: using few-shot prompting with a long CoT as an example, directly prompting models to generate detailed answers with reflection and verification, gathering DeepSeek-R1-Zero outputs in a readable format, and refining the results through post-processing by human annotators.

In this work, we collect thousands of cold-start data to fine-tune the DeepSeek-V3-Base as the starting point for RL. Compared to DeepSeek-R1-Zero, the advantages of cold start data

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCodeBench
	pass@1	cons@64	pass@1	pass@1	pass@1
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9
DeepSeek-R1-Zero-Qwen-32B	47.0	60.0	91.6	55.0	40.2
DeepSeek-R1-Distill-Qwen-32B	**72.6**	**83.3**	**94.3**	**62.1**	**57.2**

Table 6 | Comparison of distilled and RL Models on Reasoning-Related Benchmarks.

Low-level improvements

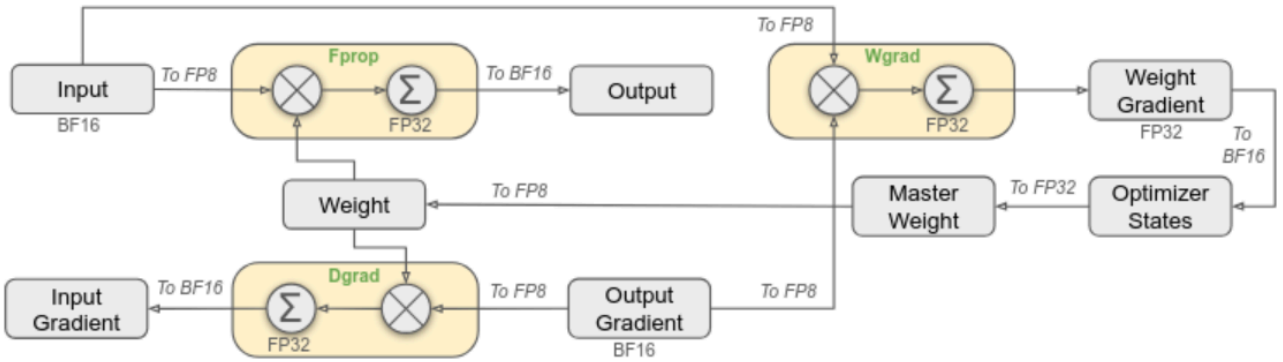


Figure 6 | The overall mixed precision framework with FP8 data format. For clarification, only the `Linear` operator is illustrated.

▶ **Is it worth developing such low-level improvements?**

What do these libraries really do ?





Error message from ArgTools

```
RequestError: HTTP/2 404 while requesting https://www.gstatic.com/devrel-devsite/prod/vf0eb6f8ebad49d1e3523c3ee6bef7563a09624802753ccc3d816c9b276e850fd/tensorflow/images/lockup.svg
```

Stack trace

Here is what happened, the most recent locations are first:

1. anonymous function
   from *Downloads.jl:285*
2. **#open#324**(f::Downloads.var"#7#8"{…}, args::String; kwargs::@Kwargs{…}) ...show types...
   from *io.jl:429*
3. **#open_nolock#1**(::Function, ::Vararg{…}; kws::@Kwargs{…}) ...show types...
   from **ArgTools** → *ArgTools.jl:35*
4. open_nolock
   from *ArgTools.jl:35*
5. **arg_write**(f::Function, arg::String)
   from **ArgTools** → *ArgTools.jl:103*

6. #download#5
   from Downloads.jl:272
7. download
   from Downloads.jl:252
8. #RobustLocalResource#30(url::String, path::String, html_attributes::Pair{…}; cache::Bool) ...show types...
   from robustlocalresource.jl:20
9. RobustLocalResource
   from robustlocalresource.jl:7
10. #save_image#2
    from Other cell: line 24

```
22 function save_image(url::URL, html_attributes...; name = split(url.url, '/')[end], kws...)
23     path = joinpath("cache", name)
24     return PlutoTeachingTools.RobustLocalResource(url.url, path, html_attributes...), path
25 end
26
```

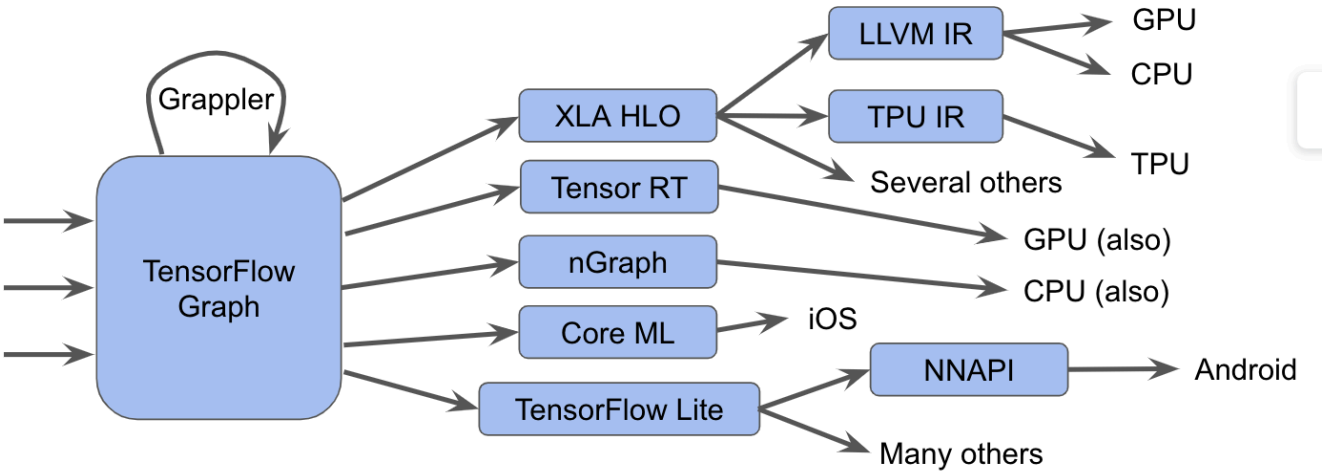
cell preview

11. Show more...

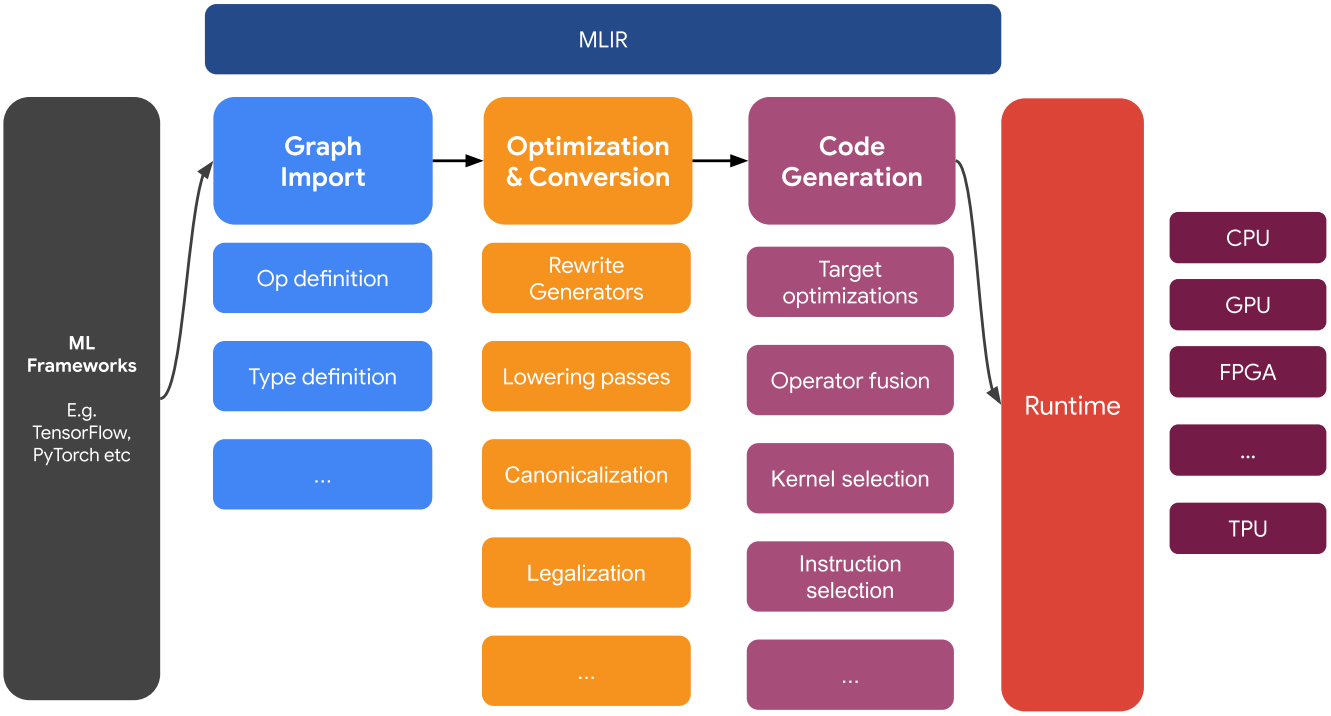
```
1 img("https://www.gstatic.com/devrel-devsite/prod/vf0eb6f8ebad49d1e3523c3ee6bef7563a09624802753ccc3d816c9b276e850fd/tensorflow/images/lockup.svg", :height => 100)
```

👀 Reading hidden code

Accelerated automatic differentation



MLIR



DIY

The goal of the course it to take a tour of **automatic differentiation**. For this we will write our own from scratch, in julia.

Yann LeCun ✓
@ylecun · **Follow**

Hotter take: ML would have advanced faster if another front-end language had been available and widely adopted instead of Python.
One that is interactive yet fast & compilable, multithreaded (no GIL), isn't bloated, doesn't care about white spaces,...
E.g. Julia or some Lisp.

> **Bojan Tunguz** ✓ @tunguz
> Hot take: Machine Learning would not have been nearly as advanced now were it not for Python. Python's two main virtues in the context of ML:
>
> 1. Lowering barriers to entry.
>
> 2. As a scripting language, it encourages and enables experimental workflow.

1:27 PM · Feb 22, 2023

3.1K **Reply** **Copy link**

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Grading

- If exam or homework is below 5, grade is the minimum of both
- If exam and homework are above 10, grade is the average of both
- Otherwise, we interpolate between these cases as follows
- Gain bonus points by contributing to the Git or winning benchmarks with the projects

```
f (generic function with 1 method)
```

```
1 f(x) = min(x/20, 0.5)
```

```
g (generic function with 1 method)
```

```
1 g(a, b) = f(a) * b + (1 - f(a)) * a
```

```
grade (generic function with 1 method)
```

```
1 grade(HW, EX) = min(g(HW, EX), g(EX, HW))
```



The End

```
import PlotlyLight
```

```
using Colors, PlutoUI, PlutoUI.ExperimentalLayout, HypertextLiteral,
PlutoTeachingTools, ShortCodes
```

```
img (generic function with 3 methods)
```

```
qa (generic function with 2 methods)
```